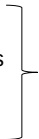


2	(a)	T_1 and T_2 : down T_3 : up	 All 3 must be correct.	B1
	(b)	By Principle of Moments, Taking moment about the pivot at the base of wire 1, sum of clockwise moments = sum of anti-clockwise moments	 $(14.5)(m_p g) = (10.0)T_3$ $T_3 = \frac{14.5}{10.0}(m_p g)$ $= \frac{14.5}{10.0}(350 \times 10^{-3})(9.81)$ $= 4.98 \text{ N}$	M1 A1
	(c)	P is in vertical translational equilibrium, R adds additional downward force. T_3 is only upward force and so must provide additional tension, more likely to snap.		B1

			If method involves principle of moments, reference to a pivot must be made known before a mark can be awarded together with the reasoning.	
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3	(a)		loss in GPE = gain in elastic PE	
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